

INTRODUCTION TO RESEARCH

Report

ALGEBRAIC GEOMETRY CODES

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1 Introduction

The development of algebraic geometry codes represents a significant milestone in coding theory by bridging concepts from algebraic geometry, with the theory of error-correcting codes.

Classical codes such as Reed-Solomon and Goppa codes, depend on the evaluation of polynomials over finite fields and have a well-understood structure. Reed–Solomon codes can be seen as codes constructed by evaluating rational functions that have a pole of restricted order at a specific point on the projective line $\mathbb{P}^1$, at points with coordinates in $\mathbb{F}_q$. Goppa codes are built by taking residues of certain functions at chosen points, where these functions have restrictions on their zeros and poles.

These two key ideas-controlling zeros and poles of functions, and using evaluations or residues at points-are the foundation of algebraic geometry codes.

By extending these ideas from the projective line to algebraic curves over finite fields, and using tools like the Riemann–Roch theorem and divisors, algebraic geometry codes provide a powerful way to build better error-correcting codes.

This report focuses on the definition and construction of Algebraic Geometry codes and their duals. We assume the reader has prior knowledge of basic algebraic geometry concepts, including divisors and function fields of algebraic curves, and classical codes, such as Reed-Solomon and Goppa codes, including their construction and properties. For a detailed background, one can refer to *Introduction to Coding Theory* by J.H. van Lint, *Algebraic Function Fields and Codes* by Henning Stichtenoth, and *Algebraic Geometry* by Robin Hartshorne, whose terminologies and results are followed closely in this report.

2 The Riemann-Roch Theorem

Before beginning, we will fix some notations. In the following sections, k is an algebraically closed field, and $\mathcal{X}$ is a smooth projective curve over k. A divisor of $\mathcal{X}$ is denoted by D, and the divisor of a rational function f on $\mathcal{X}$ is denoted by (f). A canonical divisor on $\mathcal{X}$ is denoted by W. The degree of a divisor D is denoted by $\deg(D)$. For a divisor D on the curve X, the vector space $\mathcal{L}(D)$ over k is defined by $\mathcal{L}(D) := \{f \in k(\mathcal{X})^* : (f) + D \geq 0\} \cup \{0\}$, where $k(\mathcal{X})^*$ denotes the the field of non-zero rational functions (the multiplicative group of the function field) on $\mathcal{X}$. This space is known as the *Riemann-Roch space*. Its dimension is denoted by $\dim_k \mathcal{L}(D) = l(D)$. For a rational function $f \in k(\mathcal{X})^*$ and a point $P \in \mathcal{X}$, we denote by $\mathrm{ord}_P(f)$, the discrete valuation of f at P. A rational differential form on a smooth projective curve $\mathcal{X}$ is denoted by ω, and the space of rational differential forms by $\Omega(\mathcal{X})$. The residue of a differential ω at a point P is denoted by $\mathrm{Res}_P(\omega)$.

We begin by stating the *Riemann-Roch* theorem, a central result in algebraic geometry and the key to constructing algebraic geometry codes. The proof of the *Riemann-Roch theorem*, and the corollaries following it, can be found in Hartshorne's book [2].

Theorem 2.1 (Riemann-Roch)**.** *Let D be a divisor on a smooth projective curve of genus g. Then, for any canonical divisor W*

$$l(D) - l(W - D) = \deg(D) - g + 1.$$

We state two important consequences of this theorem.

Corollary 2.1. *For any canonical divisor W, we have* $\deg(W) = 2g - 2$.

Corollary 2.2. *Let D be a divisor on a smooth projective curve of genus g and let* $\deg(D) > 2g-2$. *Then*

$$l(D) = \deg(D) - g + 1.$$

The term $l(W - D)$ in Theorem 2.1 can be interpreted in terms of rational differential forms. We do so by introducing a generalization of the *Riemann-Roch space* $\mathcal{L}(D)$ for rational differential forms.

Definition 2.1. Let D be a divisor on a curve $\mathcal{X}$. We define

$$\Omega(D) := \{\omega \in \Omega(\mathcal{X}) : (\omega) - D \geq 0\}$$

and we denote $\dim_k \Omega(D)$ by $\delta(D)$, called the *index of speciality* of D.

The connection between rational differential forms and non-zero rational functions is stated in the following theorem, and its proof is given in Stichtenoth's book [3].

Theorem 2.2. $\delta(D) = l(W - D)$.

3 Codes from Algebraic Curves

In this section, we will see the applications of algebraic geometry to coding theory, using the results from the previous section. Our alphabet will be $\mathbb{F}_q$, and the space $\mathcal{L}(D)$ will be considered over $\mathbb{F}_q$.

Let $\mathcal{X}$ be a smooth projective curve over $\mathbb{F}_q$. In the following, P_1,P_2,...,P_n are rational points on $\mathcal{X}$ and D is the divisor $P_1 + P_2 + \ldots + P_n$. Furthermore, G is another divisor such that $\operatorname{supp}(G) \cap \operatorname{supp}(D) = \varnothing$, where $\operatorname{supp}(D)$ denotes the support of D. The support of G also consists of rational points and

$$2g - 2 < \deg(G) < n. \tag{1}$$

We begin by defining the algebraic geometry code that generalizes Reed-Solomon codes.

Definition 3.1. The linear code $C(D, G)$ of length n over $\mathbb{F}_q$ is the image of the linear map $\alpha : \mathcal{L}(G) \to \mathbb{F}_q^n$ defined by $\alpha(f) := (f(P_1), f(P_2), \ldots, f(P_n))$. These are known as *geometric generalized Reed-Solomon* codes.

The following theorem states the parameters of such a code.

Theorem 3.1. *The code $C(D, G)$ has dimension $k = \deg(G) - g + 1$ and minimum distance $d \geq n - \deg(G)$.*

Proof. Recall from Definition 3.1 that $C(D, G)$ is the image of the map $\alpha\colon \mathcal{L}(G) \to \mathbb{F}_q^n$. Thus $\dim C(D, G) = \dim \mathcal{L}(G) - \dim \ker \alpha$. Let $f \in \ker \alpha$. Then $f \in \mathcal{L}(G)$ and $f(P_i) = 0$ for all i, so $\operatorname{ord}_{P_i}(f) \geq 1 = \operatorname{ord}_{P_i}(D)$. Since $\operatorname{supp}(G) \cap \operatorname{supp}(D) = \varnothing$, we have $\operatorname{ord}_{P_i}(G) = 0$. Thus at each P_i, $\operatorname{ord}_{P_i}\big((f) + G - D\big) \geq 0$. For $Q \notin \{P_1, \ldots, P_n\}$, $\operatorname{ord}_Q(D) = 0$, and $f \in \mathcal{L}(G)$ implies $\operatorname{ord}_Q\big((f) + G\big) \geq 0$, hence $\operatorname{ord}_Q\big((f) + G - D\big) \geq 0$. Therefore $(f) + G - D \geq 0$, i.e., $f \in \mathcal{L}(G - D)$, so $\ker \alpha \subset \mathcal{L}(G - D)$. Conversely, if $f \in \mathcal{L}(G - D)$, then at each P_i, $\operatorname{ord}_{P_i}(f) + \operatorname{ord}_{P_i}(G) - 1 \geq 0$, and since $\operatorname{ord}_{P_i}(G) = 0$, we get $\operatorname{ord}_{P_i}(f) \geq 1$, so $f(P_i) = 0$. Thus $f \in \ker \alpha$, giving $\mathcal{L}(G - D) \subset \ker \alpha$. Hence $\ker \alpha = \mathcal{L}(G - D)$.

Now $\deg(G-D) = \deg(G) - n < 0$ (because $\deg(G) < n$, from (1)), so $\mathcal{L}(G-D) = \{0\}$. Thus α is injective, and $\dim C(D,G) = \dim \mathcal{L}(G) = l(G)$. Since $\deg(G) > 2g-2$, by Corollary 2.2 we have $l(G) = \deg(G) - g + 1$. Therefore

$$k = \deg(G) - g + 1.$$

Let d be the minimum distance of $C(D,G)$. Then there exists a nonzero $f \in \mathcal{L}(G)$ such that the codeword $\alpha(f)$ has Hamming weight exactly d. This means f vanishes at exactly $n-d$ of the points $P_1, \ldots, P_n$. We denote these points by $P_{i_1}, \ldots, P_{i_{n-d}}$ and set $E = P_{i_1} + \cdots + P_{i_{n-d}}$. Since $f(P_{i_j}) = 0$, we have $\mathrm{ord}_{P_{i_j}}(f) \geq 1 = \mathrm{ord}_{P_{i_j}}(E)$. Because $\mathrm{supp}(G) \cap \mathrm{supp}(D) = \varnothing$ and $E \leq D$, we also have $\mathrm{supp}(G) \cap \mathrm{supp}(E) = \varnothing$. Arguing exactly as in the dimension part (pointwise), we obtain $(f) + G - E \geq 0$, so $f \in \mathcal{L}(G-E)$. Since $f \neq 0$, this implies $\mathcal{L}(G-E) \neq \{0\}$, hence $\ell(G-E) \geq 1$. Thus $\deg(G-E) \geq 0 \implies \deg(G) - (n-d) \geq 0$. Upon rearranging, we get

$$d \geq n - \deg(G).$$

□

Next, we define the algebraic geometry codes that generalize Goppa codes.

Definition 3.2. The linear code $C^*(D,G)$ of length n over $\mathbb{F}_q$ is the image of the linear map $\alpha^* : \Omega(G-D) \to \mathbb{F}_q^n$ defined by

$$\alpha^*(\eta) := (\mathrm{Res}_{P_1}(\eta), \mathrm{Res}_{P_2}(\eta), \ldots, \mathrm{Res}_{P_n}(\eta)).$$

These are known as *geometric Goppa codes*.

The parameters of this code are given by the following theorem, which we only state. A detailed proof for this theorem can be found in van Lint's book [1].

Theorem 3.2. *The code $C^*(D,G)$ has dimension $k^* = n - \deg(G) + g - 1$ and minimum distance $d^* \geq \deg(G) - 2g + 2$.*

We now prove the duality between the algebraic geometry codes $C(D,G)$ and $C^*(D,G)$. The key tool is the residue theorem, whose proof can be found in Stichtenoth's book [3] which we state, and consequently use, to prove the duality of $C(D,G)$ and $C^*(D,G)$.

Theorem 3.3 (Residue theorem)**.** *If ω is a differential on a smooth projective curve $\mathcal{X}$, then*

$$\sum_{P \in \mathcal{X}} \mathrm{Res}_P(\omega) = 0.$$

Theorem 3.4. *The codes $C(D,G)$ and $C^*(D,G)$ are dual codes.*

Proof. From Theorem 3.1 and Theorem 3.2, we know that

$$\dim C(D,G) + \dim C^*(D,G) = k + k^* = n. \tag{2}$$

It is enough to take a word from each code and show that the inner product of the two words is zero. Let $f \in \mathcal{L}(G)$ and $\eta \in \Omega(G-D)$. Since $f \in \mathcal{L}(G)$, we know that the poles of f are bounded by G, and that f is regular at $P_i \in \mathrm{supp}(D)$ since $\mathrm{supp}(G) \cap \mathrm{supp}(D) = \varnothing$. And since $\eta \in \Omega(G-D)$, the poles of η are bounded by $G-D$. It can have at most a pole of order 1 at $P_i \in \mathrm{supp}(D)$. Thus, $f\eta$ has poles of at most order 1 at $P_1, \ldots, P_n$. For any other point $Q \notin \{P_1, \ldots, P_n\}$, $f\eta$ is regular, so $\mathrm{Res}_Q(f\eta) = 0$. Hence, by Theorem 3.3, $\sum_{i=1}^n \mathrm{Res}_{P_i}(f\eta) = 0$. At P_i, we pick a local uniformizer t,

so that $t = 0$ corresponds to P_i, and expand it for f and η. We get $f = f(P_i) + a_1 t + \ldots$, since f is regular at P_i. And upon expanding η, we get $\eta = (\frac{c_{-1}}{t} + c_0 + c_1 t + \ldots)dt$, since η has at most a simple pole at P_i. Here $c_{-1} = \mathrm{Res}_{P_i}(\eta)$. Upon multiplying, $f\eta = (f(P_i) + a_1 t + \ldots)(\frac{c_{-1}}{t} + \ldots)dt$. The only term producing $1/t$ in the product is $f(P_i) \cdot \frac{c_{-1}}{t}$. Thus, $\mathrm{Res}_{P_i}(f\eta) = f(P_i) \cdot c_{-1} = f(P_i)\,\mathrm{Res}_{P_i}(\eta)$. Hence we have

$$0 = \sum_{i=1}^{n} f(P_i)\,\mathrm{Res}_{P_i}(\eta) = \langle \alpha(f), \alpha^*(\eta) \rangle.$$

Therefore, $C^*(D,G) \subseteq C(D,G)^\perp$. Together with (2), and the fact that $\dim C(D,G)^\perp = n - \dim C(D,G) = \dim C^*(D,G)$, we conclude

$$C^*(D,G) = C(D,G)^\perp.$$

□

4 An example of an Algebraic Geometry Code

In this example, we consider a code from the smooth projective Hermitian curve $\mathcal{X}$ in $\mathbb{P}^2$ over $\mathbb{F}_4$, defined by the equation $x^3 + y^3 + z^3 = 0$. In this application, k will be the algebraic closure of $\mathbb{F}_q$. The Plücker formula $g = ((d-1)(d-2))/2$ for smooth projective curves of degree d yields genus $g = 1$ for our curve $\mathcal{X}$ (since $d = 3$). Since for any $\alpha \in \mathbb{F}_4$ we have $\alpha^3 \in \{0, 1\}$, we count the rational points by cases. First, for affine points with $z = 1$, the equation becomes $x^3 + y^3 = 1$. If $x^3 = 1$ then $x \in \{1, \omega, \omega^2\}$ and $y^3 = 0$, so $y = 0$; this gives the points $P_1 = (1 : 0 : 1)$, $P_2 = (\omega : 0 : 1)$, $P_3 = (\omega^2 : 0 : 1)$. If $x^3 = 0$ then $x = 0$ and $y^3 = 1$, so $y \in \{1, \omega, \omega^2\}$; this gives $P_4 = (0 : 1 : 1)$, $P_5 = (0 : \omega : 1)$, $P_6 = (0 : \omega^2 : 1)$. For points with $z = 0$, the equation reduces to $x^3 + y^3 = 0$, i.e. $x^3 = y^3$. Since $z = 0$, not both x and y can vanish; setting $y = 1$ yields $x^3 = 1$, hence $x \in \{1, \omega, \omega^2\}$. These provide the points $P_7 = (1 : 1 : 0)$, $P_8 = (\omega : 1 : 0)$, $P_9 = (\omega^2 : 1 : 0)$. Thus $\mathcal{X}$ has nine rational points in total. We take $G := 5Q$, where $Q = (0 : 1 : 1)$. A basis of $\mathcal{L}(5Q)$ can be found. At $Q = (0 : 1 : 1)$, we take $t = x/z$ as a local parameter (since $x = 0$ at Q, and the curve is smooth there). Although the rational function $f := x/(y+z)$ is defined on most of $\mathcal{X}$, it is not regular at Q because $y + z$ vanishes there (since characteristic of field $\mathbb{F}_4$ is 2). To determine its behavior near Q, we rewrite it using the curve equation, and get $x/(y+z) = x(y^2 + yz + z^2)/(y^3 + z^3) = t^{-2} \cdot (y^2 + yz + z^2)/(z^2)$. In terms of the local uniformizer $t = x/z$ (and scaling $z = 1$ near Q), we have $x = t$, and $y^2 + yz + z^2$ is regular and non-zero at Q. Thus $x/(y+z)$ has a pole of order 2 at Q. Similarly, $y/(y+z)$ has a pole of order 3 in Q. So the pole order of $f_{i,j} = x^i y^j/(y+z)^{i+j}$ at Q is $2i + 3j$. For $f_{i,j}$ to belong to $\mathcal{L}(5Q)$, we must have $2i + 3j \leq 5$. The admissible pairs for (i, j) are $(0, 0)$, $(1, 0)$, $(2, 0)$, $(0, 1)$, and $(1, 1)$, yielding five functions. The functions $f_{0,0} = 1$, $f_{1,0} = x/(y+z)$, $f_{2,0} = x^2/(y+z)^2$, $f_{0,1} = y/(y+z)$, $f_{1,1} = xy/(y+z)^2$ have distinct pole orders $0, 2, 4, 3, 5$ respectively at Q. Since functions with distinct pole orders at a point are linearly independent, the five functions form a basis of $\mathcal{L}(5Q)$, which matches $l(5Q) = 5$ by Corollary 2.2. Next we find a basis of $C(D, 5Q)$. We take D to be the sum of the eight rational points other than $Q = P_4$. The code $C(D, 5Q)$ is spanned by the evaluation of the five functions $f_{0,0}, f_{1,0}, f_{2,0}, f_{0,1}, f_{1,1}$ at the eight points of D. Explicitly, the basis codewords are $\mathbf{c}_{0,0} = (1, 1, 1, 1, 1, 1, 1, 1)$, $\mathbf{c}_{1,0} = (1, \omega, \omega^2, 0, 0, 1, \omega, \omega^2)$, $\mathbf{c}_{2,0} = (1, \omega^2, \omega, 0, 0, 1, \omega^2, \omega)$, $\mathbf{c}_{0,1} = (0, 0, 0, \omega^2, \omega, 1, 1, 1)$, and $\mathbf{c}_{1,1} = (0, 0, 0, 0, 0, 1, \omega, \omega^2)$, listed in the order of the points $P_1, P_2, P_3, P_5, P_6, P_7, P_8, P_9$. By Theorem 3.1 the parameters of the code are $n = 8$, $k = 5$, and $d \geq 3$. The bound of Theorem 3.1 gives $d \geq 3$. Since the codeword $\mathbf{c}_{1,1}$ has weight exactly 3, the minimum distance is $d = 3$.

References

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[2] Robin Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics, Vol. 52, Springer-Verlag, New York, 1977.

[3] Henning Stichtenoth, *Algebraic Function Fields and Codes*, 2nd ed., Graduate Texts in Mathematics, Vol. 254, Springer-Verlag, Berlin, 2009.